

Cheat Sheet: Maps, Waffles, WordCloud and Seaborn

Review this cheat sheet to learn about various functions, their descriptions, syntaxes, and examples.

Folium

Map

Create a map object with specified center coordinates and zoom level.

Syntax:

```
folium.Map(location=[lat, lon], zoom_start=n)
```

Example:

```
world_map = folium.Map()  
canada = folium.Map(location=[56.130, -106.35], zoom_start=4)
```

Visual:



Marker

Add a marker to the map with custom icon, popup, and tiles

Syntax:

```
Tiles as Stamen Toner  folium.Marker(location=[lat , lon ],
popup='Marker Popup',
tiles='Stamen Toner').add_to(map)
```

Example:

```
folium.Marker(location=[556.130, -106.35],
tooltip='Marker',
tiles='Stamen Toner').add_to(world_map)
```

Visual:



Marker

Tiles as Stamen Terrain

Syntax:

```
folium.Marker(location=[lat , lon ],  
popup='Marker Popup',  
tiles='Stamen Terrain').add_to(map)
```

Example:

```
folium.Marker(location=[556.130, -106.35],  
tooltip='Marker',  
tiles='Stamen Terrain').add_to(world_map)
```

Visual:



Circle

Add a circle to the map with specified radius, color, and fill opacity.

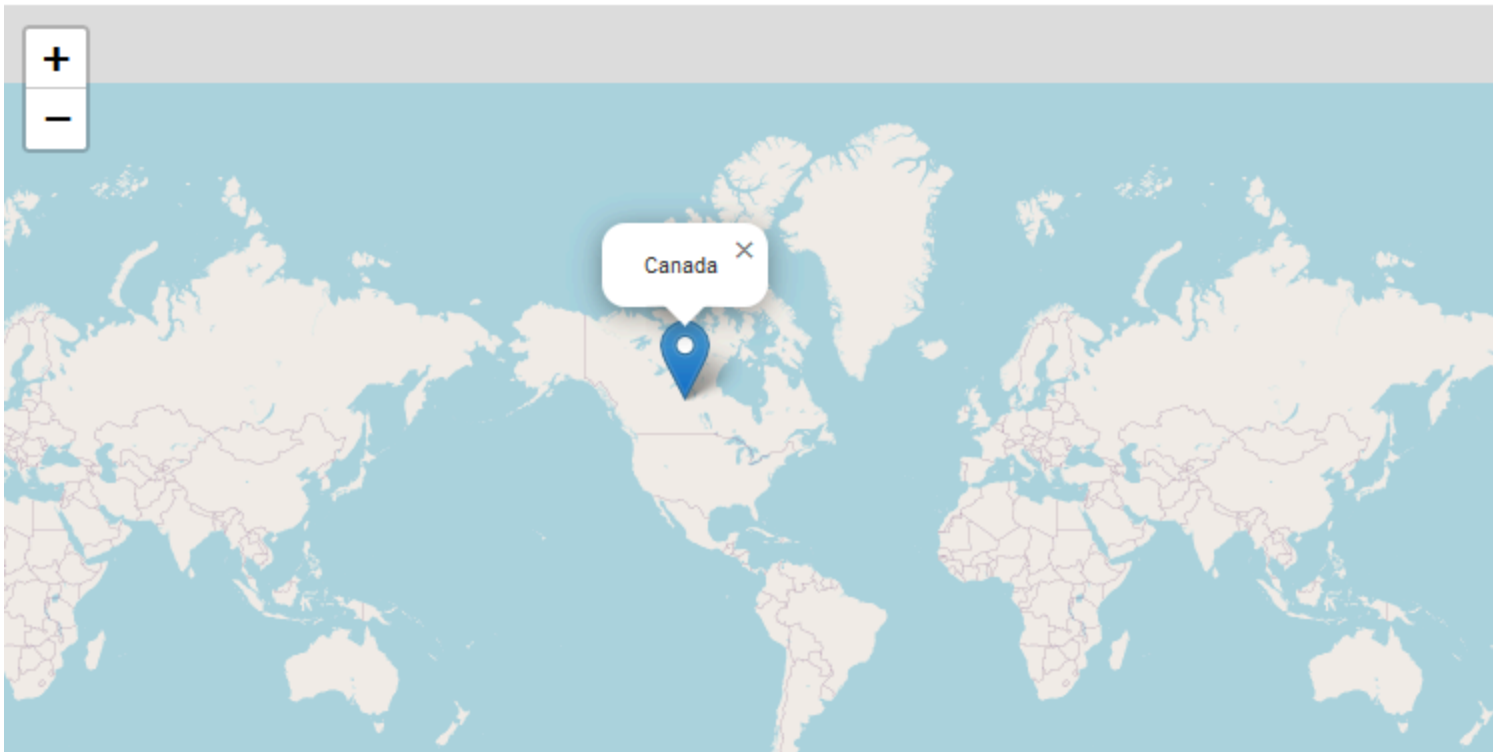
Syntax:

```
folium.features.CircleMarker(location=[lat, lon],  
radius=n, color='red',  
fill_opacity=n).add_to(map)
```

Example:

```
folium.features.CircleMarker(location=[56.130, -106.35],  
radius=1000, color='red',  
fill_opacity=0.5).add_to(world_map)
```

Visual:



Chorpleth

Create a choropleth map based on a GeoJSON file and a specified data column.

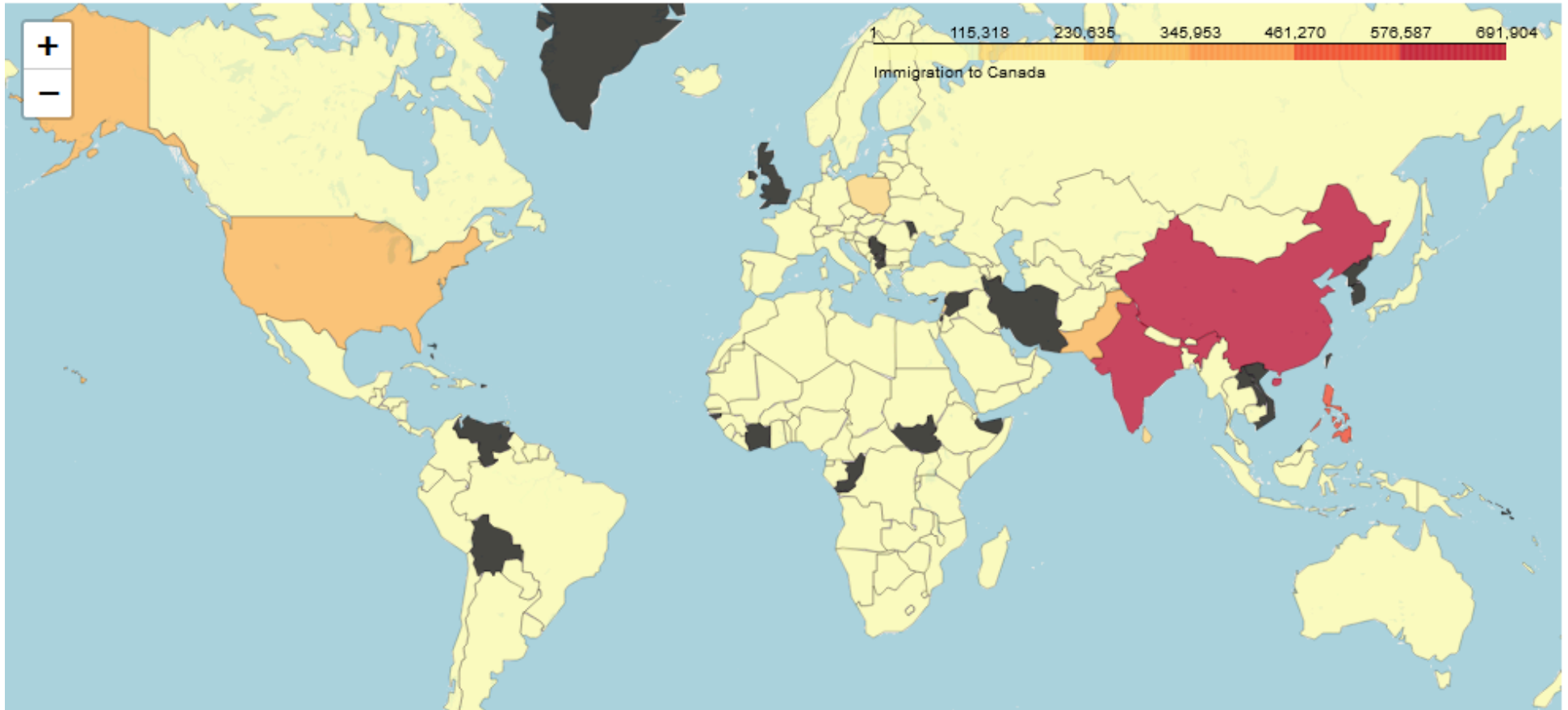
Syntax:

```
folium.Choropleth(geo_data='path/to/geojson_file',  
data=df, columns=['region', 'value_column'],  
key_on='feature.properties.id', fill_color='YlGnBu',  
fill_opacity=0.7, line_opacity=0.2,  
legend_name='Legend').add_to(map)
```

Example:

```
world_map.choropleth(geo_data=world_geo,  
data=df_can, columns=['Country', 'Total'],  
key_on='feature.properties.name', fill_color='YlOrRd',  
fill_opacity=0.7, line_opacity=0.2,  
legend_name='Immigration to Canada')
```

Visual:



PyWaffle

Waffle

Create a waffle chart based on values and categories.

Syntax:

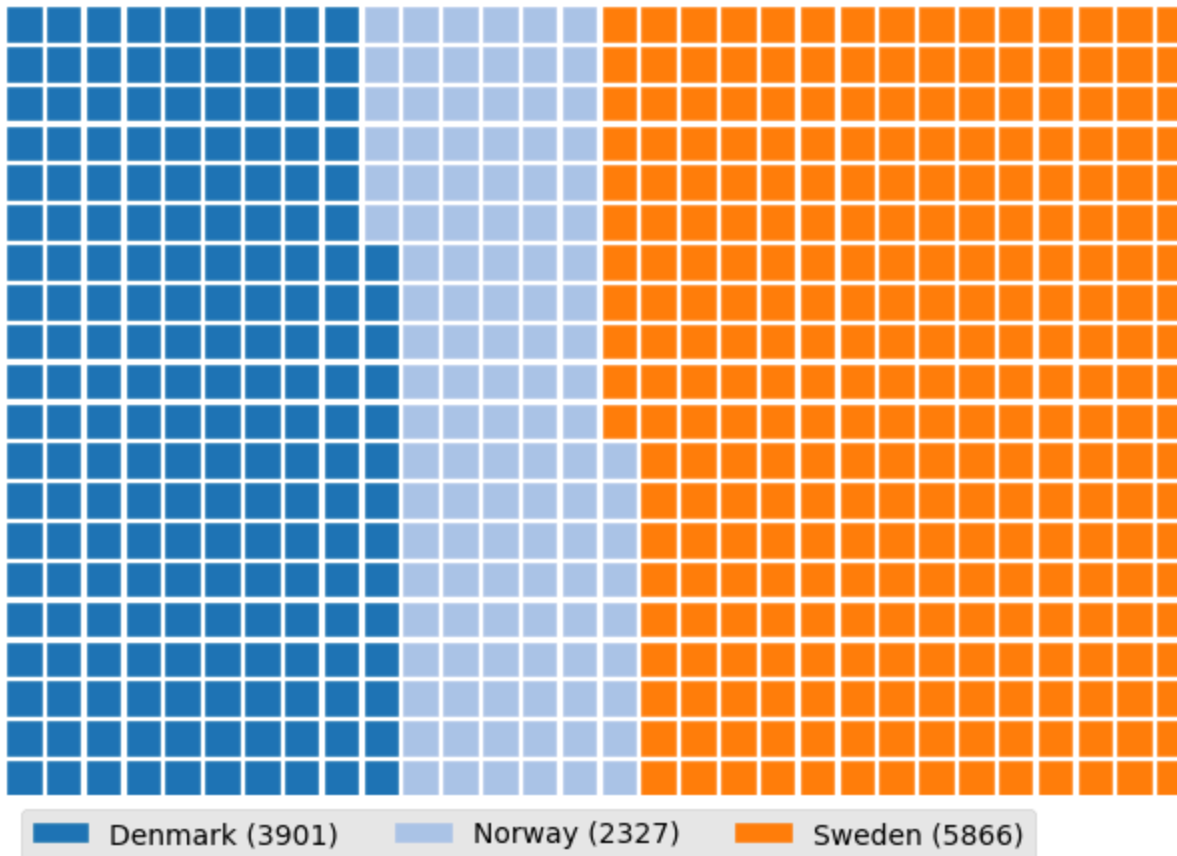
```
plt.figure(FigureClass = Waffle, rows = 20, columns = 30, values = values)
```

```
waffle_chart = waffle.Waffle(values=[value1, value2, ...],
```

Example:

```
rows=n, columns=n) plt.figure(FigureClass = Waffle,rows = 20, columns = 30,  
values = df_dsn['Total'], cmap_name = 'tab20',  
legend = {'labels': label,'loc': 'lower left',  
'bbox_to_anchor': (0,-0.1),'ncol': 3})
```

Visual:



Legend

Add a legend to the waffle chart.

Syntax:

```
waffle_chart.legend(loc='upper left', bbox_to_anchor=(1, 1))
```

Title

Add a title to the waffle chart.

Syntax:

```
waffle_chart.set_title('Waffle Chart Title')
```

Labels

Add labels to the waffle chart.

Syntax:

```
waffle_chart.set_labels(['Label 1', 'Label 2', ...])
```

WordCloud

WordCloud Create a word cloud object based on text data.

Syntax:

```
wordcloud = WordCloud().generate(text_data)
```

Example:

```
alice_wc = WordCloud(background_color='white', max_words=2000, mask=alice_mask, stopwords=stopwords)
alice_wc.generate(alice_novel)
```

```
plt.imshow(alice_wc, interpolation='bilinear')
```

Visual:





Generate the word cloud based on the text data.

```
wordcloud.generate(text_data)
```

Display the word cloud using matplotlib or other plotting libraries.

```
plt.imshow(wordcloud, interpolation='bilinear')
```

Set various options for the word cloud, such as font, colors, mask, and stopwords.

<https://www.coursera.org/learn/python-for-data-visualization/supplement/MEg5W/cheat-sheet-maps-waffles-wordcloud-and-seaborn>

```
wordcloud = WordCloud(font_path='path/to/font_file',  
background_color='white',  
colormap='Blues', mask=mask_image,  
stopwords=stopwords).generate(text_data)
```

Seaborn

barplot

Create a bar plot to visualize the relationship between a categorical variable and a numeric variable.

Syntax:

```
sns.barplot(x='x_variable', y='y_variable', data=dataframe)
```

Example:

```
sns.barplot(x='Continent', y='Total', data=df_can1)
```

Visual:

countplot

Create a count plot to display the frequency of each category in a categorical variable.

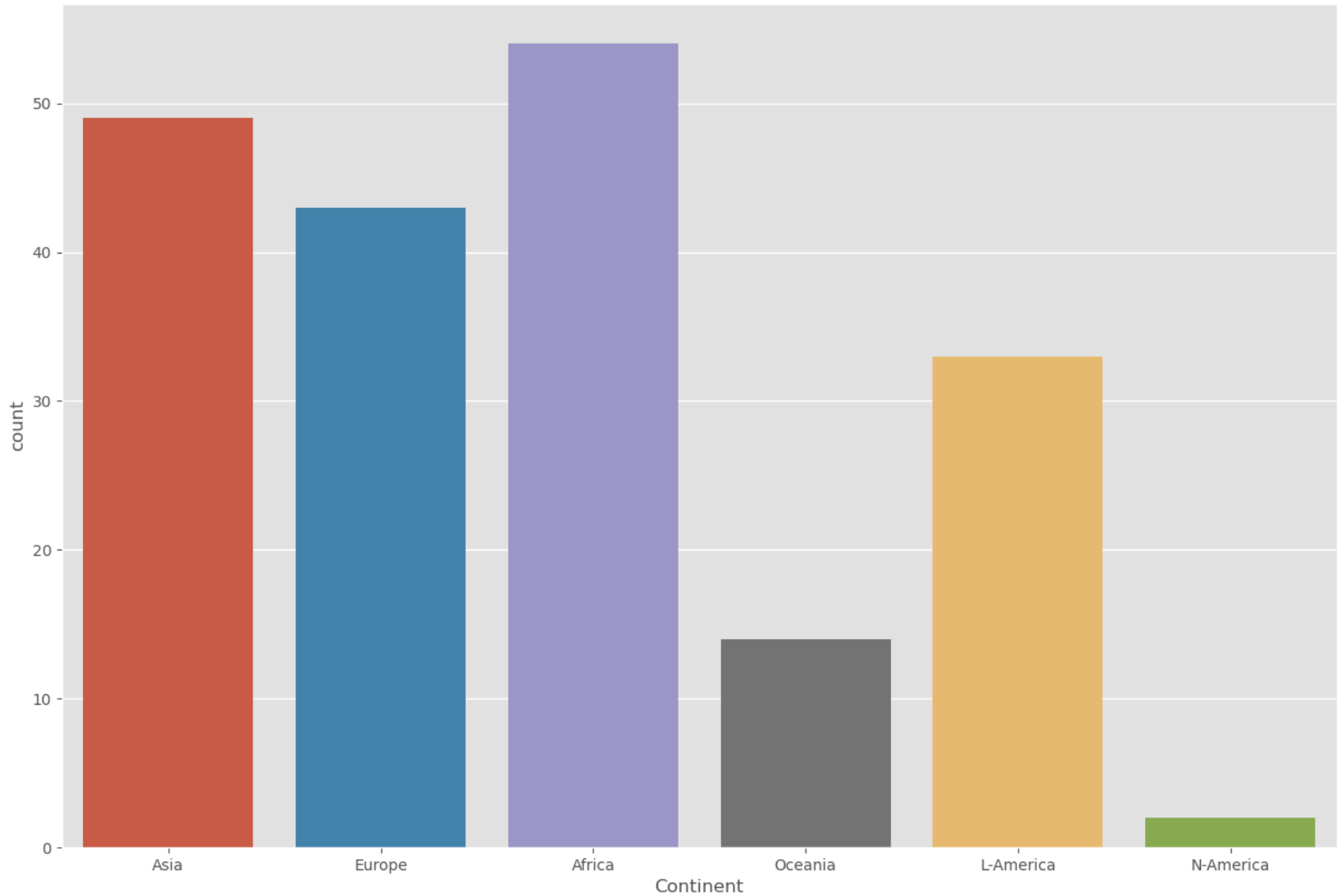
Syntax:

```
sns.countplot(x='category', data=dataframe)
```

Example:

```
sns.countplot(x='Continent', data=df_can)
```

Visual:



regplot

Create a scatter plot with a linear regression line to visualize the relationship between two numeric variables.

Syntax:

```
sns.regplot(x='x_variable',  
            y='y_variable', data=dataframe)
```

Example:

```
sns.regplot(x='year', y='total', data=df_tot)
```

Visual:

